ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science |

Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** force	Definition.
2	**(b)** force / area	Formula.
3	**(d)** particles	Tiny particles.
4	**(a)** O	O.
5	**(d)** homogeneous mixture	Uniform.
6	**(d)** smooth surfaces	Smooth.
7	**(b)** eye	Calm centre.
8	**(a)** freezing	Fusion reverse.
9	**(b)** iron	Iron.
10	**(d)** hot water	Heat helps.
11	**(d)** weight	Mass x gravity.
12	**(a)** air pressure	Atmospheric pressure.
13	**(b)** fit into the spaces	Gaps.
14	**(a)** wires	Wire drawing.
15	**(d)** mass of solute + mass of solvent	Total.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	A push or a pull (2).	2
17	Force acting per unit area (2).	2
18	Tiny particles (2).	2
19	Pure substance made of one kind of atom (2).	2
20	Homogeneous mixture of solute and solvent (2).	2

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Changes speed (1); Changes direction (1); Changes shape (1).	1+1+1
22	P = F/A (1); 40 / 2 (1); 20 Pa (1).	1+1+1
23	Fixed shape (1); Fixed volume (1); Cannot be compressed (1).	1+1+1
24	H, O (1); C, N (1); S, Ca (1).	1+1+1
	OR: Sugar and water (1); CO2 and water (1); Iodine and alcohol (1).	1+1+1
25	Stay indoors (1); Avoid trees (1); Avoid water (1).	1+1+1
	OR: Stay indoors (1); Listen to radio (1); Move to shelters (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Starts motion (1); Stops motion (1); Changes speed (1); Changes direction (1); Changes shape (1).	1+1+1+1+1
	OR: P = F/A (1); Pressure rises when area falls (1); Knife (1); Camel (1); Tractor (1).	1+1+1+1+1
27	Tiny particles (1); Spaces between them (1); Constant motion (1); Forces of attraction (1); Examples (1).	1+1+1+1+1
	OR: Composition (1); Types of atoms (1); Breaking down (1); Examples (1); Symbols/formulae (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science |

Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** newton	N.
2	**(b)** pascal	Pa.
3	**(c)** have spaces between them	Spaces present.
4	**(a)** H	H.
5	**(c)** solute	Dissolved substance.
6	**(c)** friction	It's a lubricant.
7	**(c)** wind speed	Wind.
8	**(d)** sublimation	Solid to gas.
9	**(c)** sulphur	Sulphur.
10	**(c)** faster	Mixing.
11	**(d)** at rest	Prevents motion.
12	**(d)** Atlantic	Region.
13	**(a)** diffusion	Mixing.
14	**(c)** sheets	Sheets.
15	**(b)** 100 g	Sum.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Oiling (1); Using ball bearings (1).	1+1
17	Pressure due to weight of air (2).	2
18	Liquid changing to vapour below boiling point (2).	2
19	Iron and copper (1); Sulphur and carbon (1).	1+1
20	Heterogeneous mixture whose particles settle (2).	2

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Drying clothes (1); Drying puddles (1); Sweat drying (1).	1+1+1
22	Water (1); Salt (1); Carbon dioxide (1).	1+1+1
23	Solution = 50 g (1); 10/50 x 100 (1); 20 per cent (1).	1+1+1
24	Pushing a door (1); Pulling a rope (1); Kicking a ball and drawing water (1).	1+1+1
	OR: Uproots trees (1); Damages houses (1); Cuts power lines (1).	1+1+1
25	Sweat (1); Earthen pots (1); Coolers (1).	1+1+1
	OR: Fixed composition (1); New properties (1); Separated by chemical methods (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Solute (1); Solvent (1); Sugar-water (1); CO2-water (1); Iodine-alcohol (1).	1+1+1+1+1
	OR: Attracts iron (1); Poles attract and repel (1); Acts at a distance (1); Compass (1); Made into magnets (1).	1+1+1+1+1
27	Weight of air (1); Suction cup (1); Straw (1); Syringe (1); Barometer (1).	1+1+1+1+1
	OR: Ice warms to 0 (1); Melts at 0 (1); Water warms to 100 (1); Boils at 100 (1); Steam warms (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science |

Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** move a still object	Moves things.
2	**(a)** increase pressure	Small area.
3	**(c)** fixed shape	Definite shape.
4	**(a)** C	C.
5	**(b)** solvent	Dissolver.
6	**(b)** friction	Grip.
7	**(c)** rotating funnel	Whirlwind.
8	**(b)** solids	Vibrate in position.
9	**(a)** mercury	Mercury.
10	**(c)** faster	Surface.
11	**(c)** friction	Surface effect.
12	**(a)** seriously	Danger.
13	**(b)** faster than cold water	Fast particles.
14	**(b)** lustre	Shine.
15	**(a)** temperature	Heat.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Spring balance (2).	2
17	Wind from sea to land (1); Blows during the day (1).	1+1
18	0 degrees Celsius (2).	2
19	Brass (1); Steel (1).	1+1
20	Dissolves many substances (2).	2

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Solid in liquid (1); Gas in liquid (1); Gas in gas (1).	1+1+1
22	Muscular (1); Friction (1); Push (1).	1+1+1
23	P = F/A (1); 90 / 3 (1); 30 Pa (1).	1+1+1
24	No fixed shape (1); Fixed volume (1); Flow (1).	1+1+1
	OR: Iron (1); Copper (1); Zinc (1).	1+1+1
25	Homogeneous (1); Transparent (1); Particles don't settle (1).	1+1+1
	OR: Attract iron (1); Have two poles (1); Like poles repel (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Uniform composition (1); Non-uniform (1); Salt water, air, brass (1); Sand-water, oil-water, soil (1); Different separation methods (1).	1+1+1+1+1
	OR: Dissolves many solids (1); Dissolves gases (1); Dissolves liquids (1); Used in life (1); Not all substances (1).	1+1+1+1+1
27	Surface irregularities (1); Static (1); Sliding (1); Rolling (1); Examples (1).	1+1+1+1+1
	OR: 200/4 = 50 Pa (1); 200/0.5 = 400 Pa (1); Smaller area, larger pressure (1); Units pascal (1); Eight times (1).	1+1+1+1+1

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